

Machine Learning and its Applications

Today's goals

Module 1

Machine Learning Fundamentals I: Basic Concepts

Welcome

Welcome to the specialization - Applied AI for Engineers and Scientists: Foundations

Video • 6 min

Specialization and Course Structure

Reading • 10 min

Specialization Sample Certificate

Reading • 10 min

Stay connected with UofG Online

Reading • 5 min

Module 7 Introduction

Introduction to Module 7's Study

Video • 1 min

How to Access MATLAB Online

Reading • 5 min

Machine Learning in Engineering Practice: Tool, Replacement, or Decision Aid?

Reading • 5 min

Module 7-1 Machine Learning Fundamentals

Machine Learning Fundamentals: What is Machine Learning

Video • 6 min

Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (1)

Video • 6 min

Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (2)

Video • 8 min

Materials for Machine Learning Fundamentals

Reading • 10 min

From Explicit Rules to Learned Models: What Actually Changes?

Reading • 5 min

Module 1 Learning Coach

Dialogue • 5 min

Module 7-2 Mapping Inputs to Outputs

Mapping Inputs to Outputs: Data Representation

Video • 10 min

Mapping Inputs to Outputs: Parametric ML Model

Video • 7 min

Mapping Inputs to Outputs: Non-Parametric ML Model

Video • 7 min

Mapping Inputs to Outputs: Evaluate Output

Video • 2 min

Materials for Mapping Inputs to Outputs

Reading • 10 min

How Data Representation and Model Choice Shape What a Model Can Learn

Reading • 5 min

Module 7-3 MATLAB Implementation

MATLAB Implementation: Simple Linear Regression and KNN

Video • 6 min

Materials about MATLAB Implementation

Reading • 10 min

From Concept to Code: What the Model Is Actually Doing

Reading • 5 min

Module 7 Quiz

Module 7 Quiz

Practice Assignment • 30 min

Module 7 Assignments

Module 7 Assignment 1

Graded App Item • 30 min

Module 7 Assignment 2

Graded App Item • 30 min

CSV File for Module 7 Assignment

Reading • 1 min

Module 7 Recap

Reading • 5 min

Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

Module 3

Data Preparation

Module 4

Support Vector Machines

Module 5

Artificial Neural Networks

How to Access MATLAB Online



You'll get lots of hands-on practice using MATLAB while working through the material in this course. Follow the steps below to access MATLAB Online. Even if you already have an installed version of MATLAB, we recommend using the free MATLAB Online license since it will ensure you have access to all the features and toolboxes necessary to complete the coursework.

1. If you do not already have a MathWorks account, create one [here](#). Be sure to verify your email (check your Junk/Spam folders) before moving on to step 2.
2. Visit [this page to access the free MATLAB Online license](#). Click the blue "Access MATLAB Online" button on this page to access MATLAB.
3. (Optional) Bookmark <https://matlab.mathworks.com/> to access MATLAB Online throughout the course

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